

MILSTONE-1 DOCUMENT

VinBigData Chest X-ray Abnormality Detection

Internship Project – Module 1

1. Dataset Description

1.1 Introduction

The dataset used for this project is the **VinBigData Chest X-ray Abnormalities Detection dataset**, released as part of a large-scale medical imaging challenge.

VinBigData Chest X-ray Abnormalities Detection

This dataset focuses on the automatic detection and localization of thoracic abnormalities in chest radiographs (X-rays). The goal is to develop computer vision models capable of identifying disease patterns and marking their precise locations using bounding boxes.

Medical imaging datasets are typically complex due to:

- High image resolution
- High dynamic pixel depth
- Inter-observer variability among radiologists
- Severe class imbalance

This dataset is particularly valuable because it includes radiologist-annotated bounding boxes, making it suitable for object detection models.

1.2 Dataset Composition

The dataset consists of 18,000 postero-anterior (PA) chest X-ray scans:

- **Training Set:** 15,000 labeled images
- **Test Set:** 3,000 images (labels withheld for benchmarking)

Image Format

- Format: DICOM (.dicom)
- High bit-depth (12-bit or 16-bit grayscale)
- High resolution (often $>2000 \times 2000$ pixels)

DICOM format contains both:

- Pixel data
- Metadata (patient info, acquisition parameters, photometric interpretation)

Because deep learning frameworks expect standard image formats, preprocessing is necessary before training.

1.3 Annotation Characteristics

Each training image was annotated independently by multiple experienced radiologists. This results in:

- Multiple annotations per image
- Potential disagreement between annotators
- Real-world clinical variability

The annotations are stored in train.csv and contain:

- image_id – Unique identifier
- class_name – Name of abnormality

- `class_id` – Integer label (0–14)
- `rad_id` – Radiologist identifier
- `x_min`, `y_min`, `x_max`, `y_max` – Bounding box coordinates

For images labeled as **“No Finding”** (`class_id = 14`):

- Bounding box fields are empty
 - These images contain no visible abnormalities
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1.4 Disease Categories

The dataset includes 14 pathological findings and one normal class:

Examples include:

- Cardiomegaly
- Pleural Effusion
- Pulmonary Fibrosis
- Pneumothorax
- Nodule/Mass
- Atelectasis
- Lung Opacity
- No Finding

The dataset is highly imbalanced. The “No Finding” class dominates the dataset, while conditions like Pneumothorax and Atelectasis are relatively rare. This imbalance must be considered during model training using techniques such as:

- Class-weighted loss
- Data augmentation
- Oversampling minority classes

1.5 Dataset Challenges

Key challenges in this dataset include:

1. High resolution images (large memory usage)
2. DICOM-specific handling requirements
3. Photometric variations (MONOCHROME1 vs MONOCHROME2)
4. Class imbalance
5. Multi-annotator variability
6. Bounding box scaling after resizing

Proper preprocessing is essential to ensure the data is usable by modern object detection models.

2. Data Preprocessing

2.1 Overview

Raw medical imaging data cannot be directly used for deep learning models. A structured preprocessing pipeline was implemented to:

- Convert DICOM to standard image format
- Normalize pixel intensity
- Resize images
- Parse bounding box annotations
- Scale annotations to match resized images

This pipeline ensures compatibility with convolutional neural networks and object detection architectures.

2.2 DICOM to PNG Conversion

DICOM images contain high dynamic range pixel values and metadata. Standard computer vision libraries do not directly process DICOM format efficiently.

Therefore, the following steps were implemented:

1. Read DICOM file using pydicom
2. Extract pixel array
3. Apply min–max normalization
4. Scale pixel values to range 0–255
5. Convert to 8-bit unsigned integer
6. Save as PNG format

This conversion ensures:

- Compatibility with OpenCV
- Reduced memory usage
- Easier visualization and debugging

2.3 Image Normalization

Medical images often contain pixel values exceeding 2000 due to higher bit depth.

To standardize input for neural networks:

$$\text{Normalized Pixel} = \frac{\text{Pixel} - \text{Min}}{\text{Max} - \text{Min}}$$

This scales pixel values to range [0,1].

Normalization:

- Improves training stability
 - Prevents gradient explosion
 - Standardizes intensity distribution
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2.4 Image Resizing

Original DICOM images are very large (2000×2000+). Training deep learning models on such large images is computationally expensive.

Therefore, images were resized to a fixed dimension (e.g., 512×512).

Resizing ensures:

- Uniform model input size
- Faster training
- Reduced GPU memory usage

$$scale_x = \frac{new\ width}{original\ width}$$
$$scale_y = \frac{new\ height}{original\ height}$$

These factors are later used to adjust bounding boxes.

2.5 Annotation Parsing

The train.csv file was grouped by image_id to retrieve all annotations corresponding to each image.

Steps performed:

- Filter rows where class_id != 14

- Extract bounding box coordinates
- Store class labels
- Handle images with zero abnormalities

This ensures that only meaningful disease annotations are used for object detection.

2.6 Bounding Box Scaling

Because images were resized, original bounding box coordinates must be adjusted.

Each coordinate is scaled as:

$$x_{new} = x_{original} \times scale_x$$

$$y_{new} = y_{original} \times scale_y$$

Without scaling, bounding boxes would not align with resized images.

2.7 Verification

To validate preprocessing correctness:

- Sample images were visualized
- Bounding boxes were drawn using OpenCV
- Visual inspection confirmed correct alignment
- Images labeled “No Finding” correctly returned zero bounding boxes

This verification step ensures the integrity of the preprocessing pipeline.

3. Conclusion

The Module-1 preprocessing pipeline successfully transforms raw VinBigData DICOM images into a structured, model-ready dataset.

The following tasks were completed:

- DICOM to PNG conversion
- Pixel normalization
- Image resizing
- Annotation parsing
- Bounding box scaling
- Verification through visualization

This processed dataset is now suitable for training object detection models such as:

- Faster R-CNN
- YOLO
- RetinaNet
- EfficientDet

The preprocessing stage forms the foundation for accurate and efficient deep learning model development.
